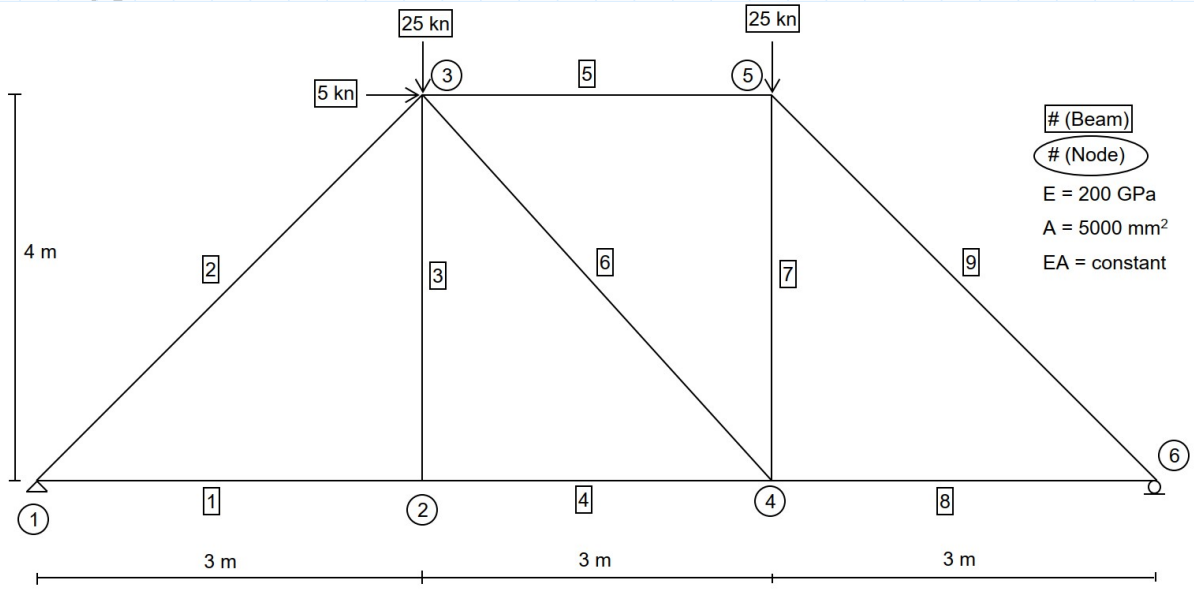
Below is the truss I have designed, where node 1 has a pinned support and node 6 has a roller support.



First, get the reaction forces

$\Sigma F_x = 0 = 50\ kN + R_{1x}$ $\qquad R_{1x} := -50\ kN$

$\Sigma M_1 = 0 = (9\ m)\ R_{6y} - (6\ m)\ (250\ kN) - (3\ m)\ (250\ kN) - (4\ m) \cdot (50\ kN)$

$$R_{6y} := \frac{(6\ m)\ (250\ kN) + (3\ m)\ (250\ kN) + (4\ m)\ (50\ kN)}{9\ m} = 272.222\ kN$$

$\Sigma F_y = 0 = R_{1y} + R_{6y} - 250\ kN - 250\ kN$

$R_{1y} := 250\ kN + 250\ kN - R_{6y} = 227.778\ kN$

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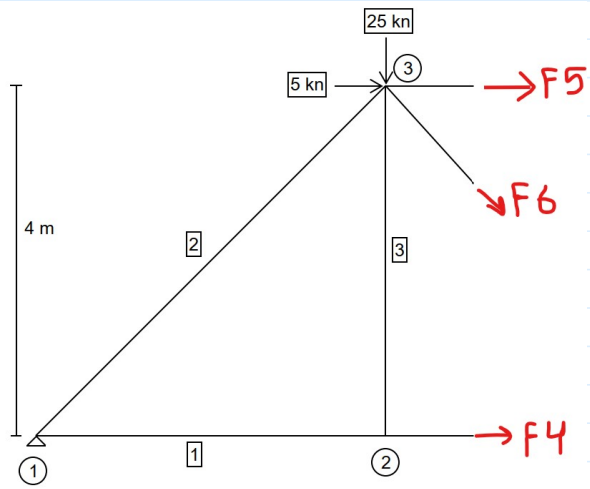
I will be using method of section to get F4 and F5 since that is easier:

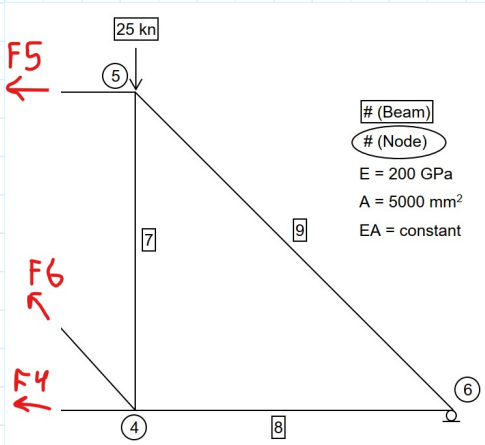


Adding negative to R1x to keeps signs right

$R_{1x} = -50\ kN$

$$\Sigma M_3 = 0 = F_4 \cdot (4\ m) - R_{1x} \cdot (4\ m) - R_{1y} \cdot (3\ m)$$

$$F_4 := \frac{(-R_{1x}) \cdot (4\ m) + R_{1y} \cdot (3\ m)}{4\ m} = 220.833\ kN$$ Tension



$$\Sigma M_4 = 0 = F_5 \cdot (4\ m) + R_{6y} (3\ m)$$

$$F_5 := \frac{-R_{6y} \cdot (3\ m)}{4\ m} = -204.167\ kN$$ Compression

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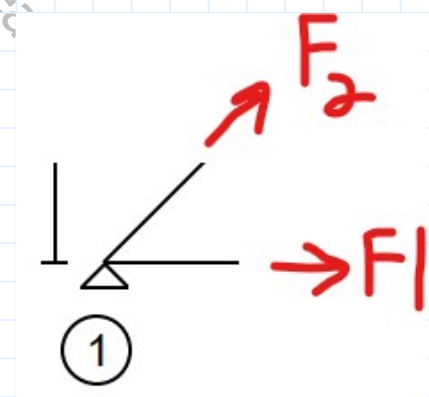
Now using Method of Joints:



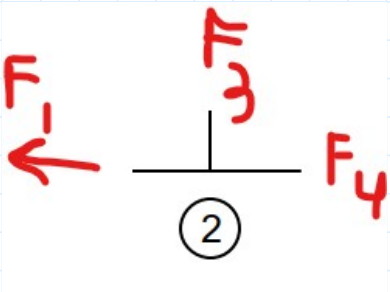
$R_{1y} = 227.778\ kN$ $\qquad R_{1x} = -50\ kN$

$\Sigma F_y = 0 = R_{1y} + \frac{4}{5} F_2$

$F_2 := -R_{1y} \cdot \left(\frac{5}{4}\right) = -284.722\ kN$ Compression

$\Sigma F_x = 0 = R_{1x} + F_1 + \frac{3}{5} F_2$

$F_1 := -R_{1x} - \frac{3}{5} F_2 = 220.833\ kN$ Tension



No external vertical load at Node 2, so F3 = 0

$F_3 := 0\ kN$

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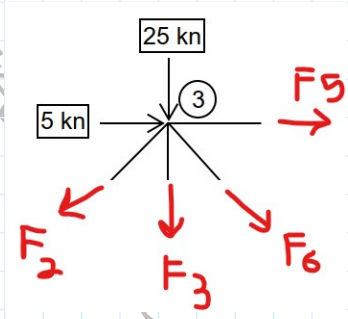
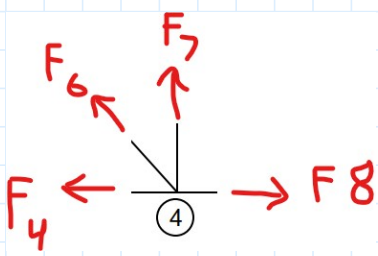


$$\Sigma F_y = 0 = -\left(\frac{4}{5}\right) F_2 - F_3 - \left(\frac{4}{5}\right) \cdot F_6 - 250\ kN$$

$$F_6 := \frac{-\left(\frac{4}{5}\right) \cdot F_2 - F_3 - 250\ kN}{\left(\frac{4}{5}\right)} = -27.778\ kN$$ Compression



$$\Sigma F_x = 0 = -F_4 - F_6 \cdot \left(\frac{3}{5}\right) + F_8$$

$$F_8 := F_4 + F_6 \cdot \left(\frac{3}{5}\right) = 204.167\ kN$$ Tension

$$\Sigma F_y = 0 = F_6 \cdot \left(\frac{4}{5}\right) + F_7$$

$$F_7 := -F_6 \cdot \left(\frac{4}{5}\right) = 22.222\ kN$$ Tension

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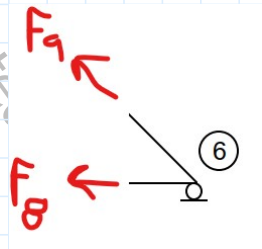

$\Sigma F_x = 0 = -F_8 - F_9 \cdot \left(\frac{3}{5}\right)$

$F_9 := -F_8 \cdot \frac{5}{3} = -340.278\ kN$ Compression

Summary:

$R_{1x} = -50\ kN$ $R_{1y} = 227.778\ kN$ $R_{6y} = 272.222\ kN$

$F_1 = 220.833\ kN$ Tension

$F_2 = -284.722\ kN$ Compression

$F_3 = 0\ kN$

$F_4 = 220.833\ kN$ Tension

$F_5 = -204.167\ kN$ Compression

$F_6 = -27.778\ kN$ Compression

$F_7 = 22.222\ kN$ Tension

$F_8 = 204.167\ kN$ Tension

$F_9 = -340.278\ kN$ Compression

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